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JupyterLab Python 3 (ipykernel)

```
[3]: #Single Inheritance - Student Result

print("----- Single Inheritance -----")
class Student:
    def details(self):
        print("Student Name : vijay")
class Result(Student):
    def marks(self):
        print("Total Marks : 485")
s = Result()
s.details()
s.marks()
```

```
----- Single Inheritance -----
Student Name : vijay
Total Marks : 485
```

```
[4]: #Multiple Inheritance - Smart Home

print("-----Multiple Inheritance-----")
class Light:
    def light_on(self):
        print("Light is ON")
class Fan:
    def fan_on(self):
        print("Fan is ON")
class SmartHome(Light,Fan):
    def status(self):
        print("Smart Home Activated")
h = SmartHome()
h.status()
h.light_on()
h.fan_on()
```

```
print("Fan is ON")
class SmartHome(light,Fan):
    def status(self):
        print("Smart Home Activated")
h = SmartHome()
h.status()
h.light_on()
h.fan_on()
```

```
-----Multiple Inheritance-----
Smart Home Activated
Light is ON
Fan is ON
```

[9]: *#Multiple Inheritance - Online Shopping*

```
print("-----Multiple Inheritance-----")
class Product:
    def price(self):
        self.amount = 5000

class Offer(Product):
    def discount(self):
        self.dis = 500

class FinalPrice(Offer):
    def bill(self):
        self.price()
        self.discount()
        print("Final Price :", self.amount - self.dis)

f = FinalPrice()
f.bill()
```

```
-----Multiple Inheritance-----
Final Price : 4500
```

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Final Price : 4500

```
[11]: #Hierarchical Inheritance - Banking System

print("-----Hierarchical Inheritance-----")
class Bank:
    def bank_name(self):
        print("BOI Bank")
class Savings(Bank):
    def interest(self):
        print("Savings Interest : 6%")
class Current(Bank):
    def overdraft(self):
        print("Overdraft Facility Available")

s = Savings()
c = Current()
s.bank_name()
s.interest()
c.bank_name()
c.overdraft()

-----Hierarchical Inheritance-----
BOI Bank
Savings Interest : 6%
BOI Bank
Overdraft Facility Available
```

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```
[12]: #Hybrid Inheritance - Hospital System

print("-----Hybrid Inheritance-----")
class person:
    def person(self):
        self.name = "vijay"
class Doctor(person):
    def doctor(self):
        self.specialist = "Cardiologist"
class Patient(person):
    def patient(self):
        self.problem = "Heart Checkup"
class Hospital(Doctor, Patient):
    def display(self):
        self.person()
        self.doctor()
        self.patient()
        print("Name :", self.name)
        print("Doctor :", self.specialist)
        print("Problem :", self.problem)
h = Hospital()
h.display()

-----Hybrid Inheritance-----
Name : vijay
Doctor : Cardiologist
Problem : Heart Checkup

[ ]:
```